GATE ME 90 Days Challenge

Day 24/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Weekly Revision	**Week 4 Revision**	**84 days to GATE**

How to use this set: 15 questions (5 MCQ, 10 NAT) on all five topics of Week 4: governors, balancing and gyroscope; free vibration; quantitative aptitude I; forced vibration; static and fatigue design (3 questions each). Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 NAT

A Watt governor runs at 60 rpm. Its height (vertical distance from the plane of rotation of the balls to the point where the arms meet the spindle axis), in mm, round off to one decimal place, is _____

Q2 NAT

The rotor of an aeroplane engine has a mass of 400 kg and a radius of gyration of 0.3 m, and spins at 3000 rpm. The aeroplane takes a turn of 100 m radius at 360 km/h. The gyroscopic couple on the aircraft (in kN·m, round off to two decimal places) is _____

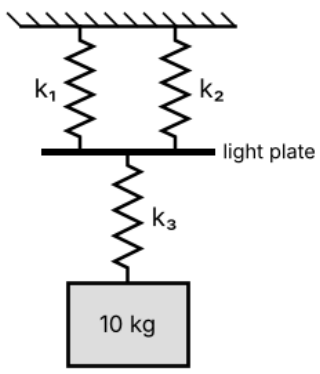
Q3 MCQ

For a reciprocating mass m with crank radius r, crank angle θ, angular speed ω and obliquity ratio n = l/r, the secondary unbalanced force is

(A) m ω² r cos 2θ
(B) m ω² r cos θ/n
(C) m ω² r cos 2θ/n
(D) m ω² r cos θ

Q4 NAT

In the system shown, springs k_1 = 2000 N/m and k_2 = 3000 N/m act side by side between the support and a light rigid plate, and spring k_3 = 5000 N/m connects the plate to a 10 kg mass. The natural frequency of vertical vibration (in Hz, round off to three decimal places) is _____



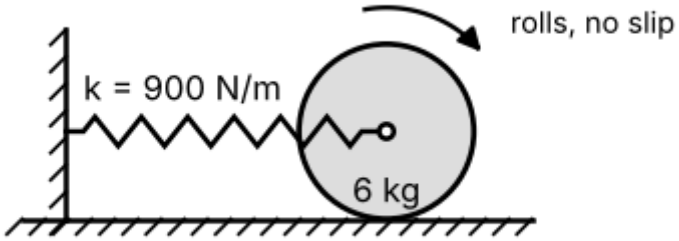
Q5 MCQ

The critical damping coefficient c_c of a spring-mass system (mass m, stiffness k, natural frequency ω_n) is

(A) $2k/\omega_n^2$
(B) $\sqrt{(km)} = m\omega_n$
(C) $2\sqrt{(km)} = 2m\omega_n$
(D) $\sqrt{(k/m)}$

Q6 NAT

A solid cylinder of mass 6 kg rolls without slipping on a horizontal floor. Its centre is attached to a horizontal spring of stiffness 900 N/m, as shown. The natural circular frequency of small oscillations (in rad/s) is _____



Q7 NAT

A car goes from P to Q at 40 km/h and returns from Q to P along the same road at 60 km/h. Its average speed for the whole journey (in km/h) is _____

Q8 MCQ

If A : B = 2 : 3 and B : C = 4 : 5, then A : B : C is

(A) 8 : 12 : 15
(B) 2 : 3 : 5
(C) 8 : 12 : 5
(D) 2 : 12 : 15

Q9 NAT

By selling an article for ₹540, a trader loses 10 %. At what price (in ₹) should he sell it to gain 15 %?

_____

Q10 NAT

A lightly damped system ($\zeta = 0.08$) is excited at resonance by a harmonic force whose amplitude would produce a static deflection of 2 mm. The resonant amplitude (in mm, round off to one decimal place) is _____

Q11 MCQ

A disc is mounted at the middle of a light shaft between bearings. The static deflection of the shaft under the weight of the disc is 0.5 mm. The critical (whirling) speed of the shaft is about

(A) 2675 rpm
(B) 946 rpm
(C) 140 rpm
(D) 1338 rpm

Q12 NAT

A machine on spring-damper mounts ($\zeta = 0.2$) runs at twice the natural frequency of the mounting. The force transmissibility (round off to three decimal places) is _____

Q13 NAT

A component has σ_m = 80 MPa and σ_a = 60 MPa. Its corrected endurance limit is 180 MPa and S_{yt} = 320 MPa. The factor of safety by the Soderberg criterion (round off to two decimal places) is _____

Q14 MCQ

Which theory of failure is most suitable for designing a component made of grey cast iron?

(A) Maximum strain energy theory
(B) Maximum shear stress theory
(C) Maximum principal stress theory
(D) Maximum distortion energy theory

Q15 NAT

The principal stresses at a point in a ductile steel part are σ_1 = 150 MPa, σ_2 = -50 MPa and σ_3 = 0. For S_{yt} = 300 MPa, the factor of safety by the maximum shear stress theory (round off to two decimal places) is _____

Answer Key

Q	Type	Answer
Q1	NAT	248.6 (accept 247.1 to 250.1)
Q2	NAT	11.31 (accept 11.24 to 11.38)
Q3	MCQ	(C) m ω² r cos 2θ/n
Q4	NAT	2.516 (accept 2.501 to 2.532)
Q5	MCQ	(C) $2\sqrt{(km)} = 2m\omega_n$
Q6	NAT	10
Q7	NAT	48
Q8	MCQ	(A) 8 : 12 : 15
Q9	NAT	690
Q10	NAT	12.5
Q11	MCQ	(D) 1338 rpm
Q12	NAT	0.412 (accept 0.410 to 0.415)
Q13	NAT	1.71 (accept 1.70 to 1.72)
Q14	MCQ	(C) Maximum principal stress theory
Q15	NAT	1.5

Detailed Solutions

Q1 **Answer: 248.6 (accept 247.1 to 250.1)**

Concept: For a Watt governor, tan θ = ω²r/g gives h = g/ω² = 895/N² m (N in rpm).

h = 895/60² = 895/3600 = **0.2486 m = 248.6 mm**

Exam tip: h ∝ 1/N², so the Watt governor is useful only at low speeds; at 300 rpm h is barely 10 mm.

Q2 **Answer: 11.31 (accept 11.24 to 11.38)**

Concept: $C = I\omega\omega_p$, with $\omega_p = v/R$ for a turn.

I = mk² = 400 × 0.3² = **36 kg·m²**; ω = 2π × 3000/60 = **314.16 rad/s**

v = 360/3.6 = 100 m/s; ω_p = 100/100 = **1 rad/s**

C = 36 × 314.16 × 1 = 11310 N·m = **11.31 kN·m**

Exam tip: Convert km/h to m/s and rpm to rad/s before multiplying; most wrong answers in GATE come from units.

Q3 **Answer: (C) m ω² r cos 2θ/n**

Concept: The inertia force of the reciprocating parts ≈ mω²r(cos θ + cos 2θ/n).

Primary part: mω²r cos θ (varies at crank speed). Secondary part: **mω²r cos 2θ/n** (varies at twice the crank speed, smaller by the factor n).

Exam tip: The secondary force acts like an imaginary crank of radius r/(4n) rotating at 2ω, which is how secondary balancing of multi-cylinder engines is done.

Q4 **Answer: 2.516 (accept 2.501 to 2.532)**

Concept: Springs sharing the same deflection are in parallel (k add); springs carrying the same force are in series (1/k add).

$k_1 \parallel k_2 = 2000 + 3000 =$ **5000 N/m**

In series with k_3: $k_e = 5000 \times 5000/(5000 + 5000) =$ **2500 N/m**

$f_n = (1/2\pi)\sqrt{(2500/10)} =$ **2.516 Hz**

Exam tip: Don't be misled by how the springs are drawn; ask which springs share the deflection and which share the force.

Q5 **Answer: (C) $2\sqrt{(km)} = 2m\omega_n$**

Concept: Critical damping is the value at which the roots of $ms^2 + cs + k = 0$ just become equal: $c^2 = 4km$.

$c_c =$ **$2\sqrt{(km)} = 2m\omega_n = 2k/\omega_n$**; damping ratio $\zeta = c/c_c$.

Exam tip: $2k/\omega_n^2$ has units of mass, not of a damping coefficient; check units to eliminate options quickly.

Q6 **Answer: 10**

Concept: Energy method. Rolling without slip: $KE = \frac{1}{2}mv^2 + \frac{1}{2}(\frac{1}{2}mR^2)(v/R)^2 = \frac{3}{4}mv^2$.

Equivalent mass = 3m/2 = 9 kg; spring stiffness at the centre = k = 900 N/m

$\omega_n = \sqrt{(k/(1.5m))} = \sqrt{(2k/3m)} = \sqrt{(1800/18)} =$ **10 rad/s**

Exam tip: If the spring were attached at the top of the cylinder, its stretch would be 2x and $\omega_n = \sqrt{(8k/3m)}$.

Q7 **Answer: 48**

Concept: For equal distances, average speed $= 2v_1v_2/(v_1 + v_2)$ (harmonic mean), not the simple average.

Average $= 2 \times 40 \times 60/(40 + 60) =$ **48 km/h**

Exam tip: 50 km/h (the arithmetic mean) is the trap option. It is correct only when equal *times* are spent at each speed.

Q8 **Answer: (A) 8 : 12 : 15**

Concept: Make the common term (B) equal in both ratios.

A : B = 2 : 3 = 8 : 12; B : C = 4 : 5 = 12 : 15

A : B : C = **8 : 12 : 15**

Exam tip: Multiply the first ratio by B's value in the second and the second ratio by B's value in the first.

Q9 **Answer: 690**

Concept: Find the cost price first; all percentages refer to CP.

CP = 540/(1 − 0.10) = **₹600**

Required SP = 600 × 1.15 = **₹690**

Exam tip: Shortcut: $SP_2 = SP_1 \times (100 + \text{gain})/(100 - \text{loss}) = 540 \times 115/90$.

Q10 **Answer: 12.5**

Concept: At r = 1, the magnification factor is 1/(2ζ).

$X = X_{st}/(2\zeta) = 2/(2 \times 0.08) =$ **12.5 mm**

Exam tip: The true peak (at $r = \sqrt{(1 - 2\zeta^2)}$) is $1/(2\zeta\sqrt{(1 - \zeta^2)})$, only about 0.3 % larger here; GATE accepts 1/(2ζ) unless it asks for the exact maximum.

Q11 **Answer: (D) 1338 rpm**

Concept: Critical speed equals the natural frequency of transverse vibration: $\omega_c = \sqrt{(k/m)} = \sqrt{(g/\delta)}$.

$\omega_c = \sqrt{(9.81/0.0005)} =$ **140.1 rad/s**

$N_c = 60 \times 140.1/2\pi =$ **1338 rpm**

Exam tip: 140 is ω in rad/s; confusing rad/s with rpm is the common trap here.

Q12 **Answer: 0.412 (accept 0.410 to 0.415)**

Concept: $T = \sqrt{(1 + (2\zeta r)^2)}/\sqrt{((1 - r^2)^2 + (2\zeta r)^2)}$.

$2\zeta r = 0.8$; $1 - r^2 = -3$

$T = \sqrt{(1 + 0.64)}/\sqrt{(9 + 0.64)} = 1.2806/3.1048 =$ **0.412**

Exam tip: Without damping T would be $1/(r^2 - 1) = 0.333$; damping raises T above $r = \sqrt{2}$.

Q13 **Answer: 1.71 (accept 1.70 to 1.72)**

Concept: Soderberg: $\sigma_a/S_e + \sigma_m/S_{yt} = 1/N$.

$1/N = 60/180 + 80/320 = 0.3333 + 0.2500 = 0.5833$

N = **1.71**

Exam tip: Soderberg also guards against yielding, so no separate yield check is needed.

Q14 **Answer: (C) Maximum principal stress theory**

Concept: Grey cast iron is brittle: it fractures without yielding when the largest normal stress reaches S_{ut}.

So the **maximum principal stress (Rankine)** theory is used. Tresca and von Mises describe *yielding* of ductile metals.

Exam tip: Brittle → Rankine (or Coulomb-Mohr when $S_{uc} \neq S_{ut}$); ductile → von Mises (or Tresca).

Q15 **Answer: 1.5**

Concept: Tresca: $\tau_{max} = (\sigma_{max} - \sigma_{min})/2$ among all three principal stresses; yield when $\tau_{max} = S_{yt}/2$.

$\sigma_{max} = 150$, $\sigma_{min} = -50$: $\tau_{max} = (150 - (-50))/2 =$ **100 MPa**

$N = (300/2)/100 =$ **1.50**

Exam tip: When σ_1 and σ_2 have the same sign, $\sigma_3 = 0$ becomes σ_{min}, and $\tau_{max} = \sigma_1/2$.

What next?

Day 24 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

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Next, Day 25/90: Machine Design, Joints. Comment **GATE90** on our reel to get all links.
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